CoReGraph: Learning to Route Graph Experts under Compositional Deployment Contracts

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Abstract

Graph systems are deployed under contracts that jointly determine time, graph visibility and construction, model-selection access, human-review budget, and compute resources. An expert that is appropriate under one contract can become invalid, unavailable, or harmful when these coordinates change together, and a global mixture cannot in general adapt when expert orderings reverse. We formulate *compositional deployment-contract generalisation*: learn from labelled source contracts and route heterogeneous experts under an unseen contract combination without target-label access. CoReGraph combines a factorised uncertainty-aware contract encoder, per-expert label-free diagnostics, a contract prior with bounded instance corrections, hard resource-feasible masks, robust contract-regret objectives, and source-calibrated abstention. We establish scoped results for fixed-mixture regret, additive-axis composition, feasible masking, and selective risk. A three-layer benchmark joins real fraud deployment shifts, fifteen controlled mechanisms, and an established graph-OOD family under common leakage and resource semantics. CLAIM_BLOCKED[MAIN-RESULT-ABSTRACT] No empirical comparison is stated until saved-output pilots, official baselines, paired statistics, and claim gates complete.

1 Introduction

Graph learning is often evaluated as if deployment were a fixed dataset split. In practice, the admissible computation changes with the operating contract. A fraud detector may move from a graph whose future structure is visible to one where target edges are isolated; a platform may replace a static snapshot with a rolling event stream; a regulator or operations team may tighten the review queue; or a graph expert may cease to fit within latency or memory limits. These interventions change not only the distribution of inputs, but also what information and actions are valid.

This distinction matters because graph structure can help or hurt. Message passing can recover relational signal when the deployment graph matches training access, yet can amplify stale, truncated, heterophilous, or unavailable structure. Feature, graph, and temporal experts can therefore reverse order across contracts. A fixed expert is unable to respond, while a global mixture gives the same mass to each expert despite a changed graph view or feasible set. Existing graph OOD benchmarks make environments explicit [5]; graph mixtures specialize computation [17, 20]; and aligned shift experts address composite graph corruptions [19]. The missing object is a typed *deployment contract* that joins distribution, information access, selection policy, budget, and resource feasibility.

We study generalisation to an unseen combination of known contract factors. The source environments may separately reveal early-to-late time, isolated graph access, or a tight review budget, while the target combines them. This is stricter than choosing among named protocols and different from unsupervised adaptation: the headline setting forbids target labels and does not assume that target covariates may be used for fitting. It is also operationally constrained. Routing to an unavailable expert is invalid even if that expert would have been accurate, and abstaining on every instance is not useful merely because selective risk becomes undefined.

We introduce CoReGraph, a family of routers for heterogeneous pretrained experts. It represents each contract axis separately, handles missing and uncertain metadata, combines label-free expert diagnostics, and forms a stable contract-level prior with a bounded instance-level correction. Availability, memory, latency, and invocation constraints are applied before normalization. A source-frozen abstention policy handles uncertainty and an empty feasible set. Training combines predictive risk with whole-contract feasible-oracle regret, worst-contract/CVaR risk, budget, stability, uncertainty, sparsity, and abstention terms.

Our contributions are:

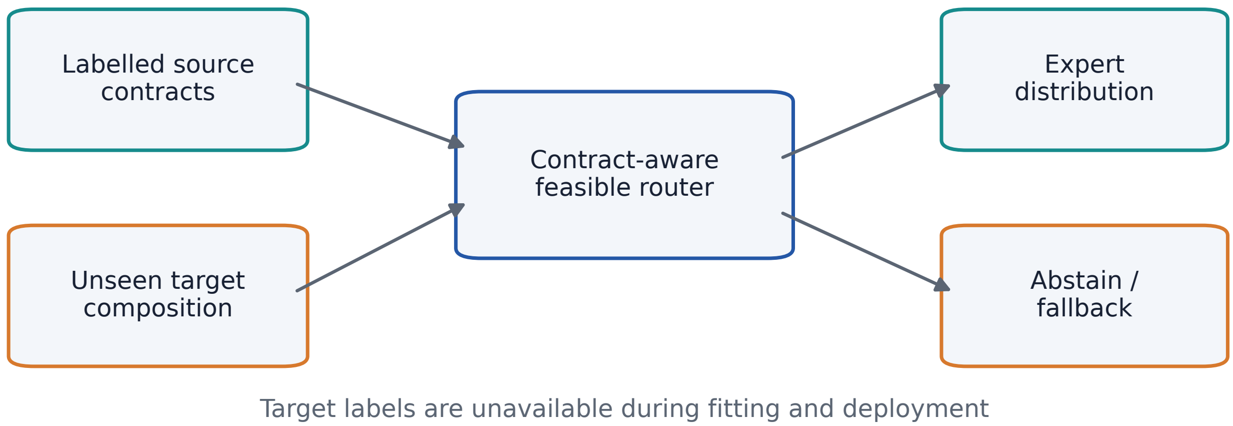

Figure 1: Compositional deployment-contract generalisation. The target combines operating conditions not jointly observed in the labelled source contracts. Target labels are unavailable during fitting and deployment.

1. a formal learning problem for compositional deployment contracts spanning time, graph visibility, graph construction, selection access, review budget, and resources;
2. a modular hierarchical router with uncertainty-aware factorisation, expert diagnostics, exact feasibility masks, robust regret training, and controlled abstention;
3. scoped theory: a fixed-mixture impossibility result, a regret decomposition, an additive-axis composition result with interaction failures, a mask guarantee, and a selective-risk transfer result;
4. a three-layer evaluation joining FraudShiftBench inputs, fifteen controlled mechanisms, and a recognized non-fraud graph-OOD family, including noisy contracts and resource interventions; and
5. a pre-registered, checksum-first, scenario-local reproducibility system that blocks every numerical claim until its evidence and statistical predicate pass.

FraudShiftBench is inherited evaluation infrastructure, not the method contribution of this paper. Its scientific files remain frozen, and its prior result tables do not support CoReGraph. The present paper is intentionally results-blocked: completing method and protocol prose before execution prevents target outcomes from silently shaping the story.

2 Related work

Graph OOD generalisation and adaptation. GOOD provides explicit covariate and concept-shift environments for node and graph tasks [5]. CIGA learns causally invariant subgraphs [1], while EERM constructs environments through graph perturbation [18]. GraphMETRO uses aligned experts for mixtures of graph shifts [19]. Domain adaptation and test-time adaptation may exploit unlabeled target inputs; our headline domain-generalisation regime forbids such access unless the selection coordinate explicitly enables it. Exact graph-OOD and adaptation coverage remains subject to official-code parity rather than an internal method inheriting an upstream name.

DomainBed shows that model selection and implementation parity can dominate apparent domain-generalisation gains [6]. Fully test-time methods such as Tent [16] and EATA [10] update a pretrained model from unlabeled test inputs. We therefore register domain generalisation, unlabeled target adaptation, and few-label adaptation as different access contracts instead of pooling them under one shift label.

Graph mixtures of experts. GMoE routes among message-passing experts to encourage specialization [17]. Mowst combines weak feature and strong graph experts using confidence [20]. These methods motivate heterogeneous

Table 1: Deployment-contract axes and uncertainty semantics.

Axis	Representative content	Non-standard state
Time	snapshot, rolling, early-to-late	missing / uncertain / unseen
Visibility	nodes, edges, target covariates	partial / out-of-support
Construction	history, orientation, topology	unseen combination
Selection	source-only, unlabeled adaptation	access typed separately
Budget	review fraction, fixed K, cost	uncertain / changing
Resource	availability, memory, latency	dynamic / empty feasible set

expert routing, but a deployment contract additionally determines whether an expert is admissible and which graph view it may consume. CoReGraph routes pretrained expert families under a typed contract and distinguishes a stable contract prior from bounded instance corrections. Confidence-only routing is a baseline and a documented failure mode.

Sparse conditional computation predates graph routing: sparsely gated MoE layers learn an input-dependent expert subset [15], and Switch Transformers expose the resulting capacity, communication, and load-balancing trade-offs at scale [3]. Our setting is smaller but more constrained: feasibility masks are deployment rules, not learned capacity decisions, and expert invocation cost is measured rather than inferred from sparsity.

Robust and invariant learning. GroupDRO optimizes worst-group risk [14]; VREx penalizes variation across environment risks [8]. We adapt robust optimization to regret relative to a whole-contract feasible expert, with mean, maximum, and CVaR aggregation. This reference changes when the resource set changes; an instance-clairvoyant oracle is reported separately and never used as a deployable comparator.

Selective prediction, budgets, and resources. Selective classification permits abstention when risk is high [4]. Our acceptance threshold is selected on source validation, frozen before target scoring, and constrained by coverage/review capacity. Resource-aware mixtures commonly trade accuracy against invocation cost; here memory, latency, availability, and invocation constraints first define a hard feasible set. We report router latency separately from expert inference and never substitute training time.

Predictive confidence can degrade under distribution shift even after in-distribution calibration [11]. This motivates reporting proper scores, coverage–risk behavior, and abstention jointly, while preventing unlabeled target diagnostics from becoming an implicit label channel.

Temporal graphs and fraud under shift. Temporal Graph Networks model timed event streams with memory [13], whereas EvolveGCN evolves convolutional parameters across graph snapshots [12]. Fraud-specific graph learning must additionally confront camouflage and neighborhood heterogeneity [2], extreme class imbalance [9], and dynamic financial graph scale [7]. Fraud graphs also pose chronological splits, delayed labels, changing structure, and human-review constraints. We preserve provider time and task semantics rather than treating every graph as static node classification. A snapshot approximation remains diagnostic unless validated against an official temporal implementation.

Compositional generalisation. Compositional learning asks whether known factors can be recombined outside observed tuples. Our factorisation is deliberately modest: it supports source-observed axis values and bounded interactions, and the theory becomes vacuous for arbitrary XOR-like effects. Unlike architectural MoE work, the composition variables here are registered deployment coordinates with typed observability and access semantics. The held-out object is a coordinate combination, not an unseen axis value, and arbitrary interaction effects remain outside the stated guarantee.

3 Problem formulation

Let a deployment contract be

$$c = (c^{\text{time}}, c^{\text{vis}}, c^{\text{build}}, c^{\text{select}}, c^{\text{budget}}, c^{\text{resource}}).$$

An axis may contain a category or continuous value together with observation state and confidence. Missing values receive an explicit mask; unseen categories map to an unknown token; out-of-range continuous values are identified

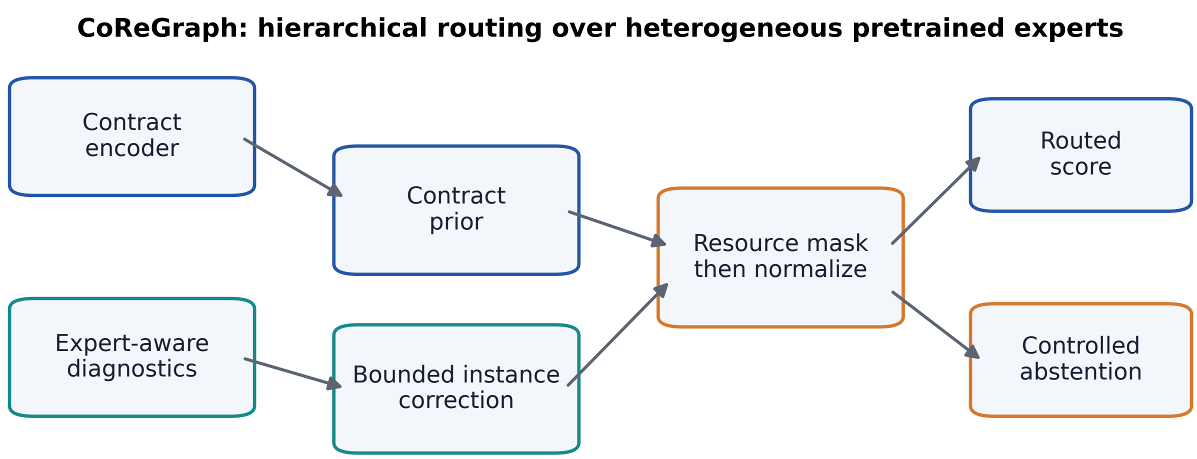

Figure 2: CoReGraph forms a contract prior, adds a bounded instance correction, imposes feasibility before normalization, and either routes or abstains. The diagram contains no empirical result.

relative to source-only normalization. An unseen *combination* contains axis values observed separately in source contracts but not their joint tuple. An unseen *value* was absent from all source contracts and is a distinct, harder regime.

Let $\mathcal{E} = \{1, \dots, E\}$ be heterogeneous experts. Contract c induces an admissible graph view G_c, prediction unit, label-access policy, review policy, and feasible set $\mathcal{A}(c) \subseteq \mathcal{E}$. Expert e emits score $s_e(x; G_c)$, label-free diagnostic vector $d_e(x, c)$, and a resource record. A router emits

$$\pi_\theta(e \mid x, c, d) \geq 0, \qquad \sum_{e \in \mathcal{A}(c)} \pi_\theta(e \mid x, c, d) = 1,$$

with zero mass outside $\mathcal{A}(c)$. If the feasible set is empty, the policy emits an explicit abstention or a declared non-expert fallback rather than a spurious distribution.

For bounded loss ℓ, contract risk is

$$R_c(\pi) = \mathbb{E}_{(X,Y)\sim P_c}\left[\ell\left(\sum_e \pi_\theta(e \mid X, c, d) s_e(X; G_c), Y\right)\right].$$

The whole-contract feasible oracle is $R_c^\star = \min_{e\in\mathcal{A}(c)} R_c(e)$. Contract regret is $\mathrm{Reg}_c(\pi) = R_c(\pi) - R_c^\star$. We report mean, maximum, and empirical CVaR regret across contracts. The oracle selects one expert only after its risk is aggregated over the contract; a per-instance clairvoyant oracle is an offline ceiling with a different estimand.

Let $a_\theta(x, c) \in \{0, 1\}$ indicate acceptance. Coverage is $P_c(a_\theta = 1)$, and selective risk is $\mathbb{E}_c[\ell \mid a_\theta = 1]$. It is undefined at zero coverage. Review budget B_c may specify a fraction, fixed K, cost matrix, or latency allowance. Resource constraints are applied before selection, while review budgets govern ranked action or abstention.

We evaluate seen contracts, unseen combinations, unseen values, partially observed and noisy contracts, dynamic resources, all-expert failure, and all-expert unavailability. Domain generalisation, unlabeled target adaptation, and few-label adaptation are separate access regimes. Target labels are unavailable for training, router selection, calibration, and threshold choice in the headline regime.

4 CoReGraph

4.1 Factorised uncertain contract representation

For axis j, the encoder combines a categorical embedding, a source-normalized continuous projection, an observation-state embedding, and confidence q_j. Missing axes use a deterministic missing token; unknown and unseen categories

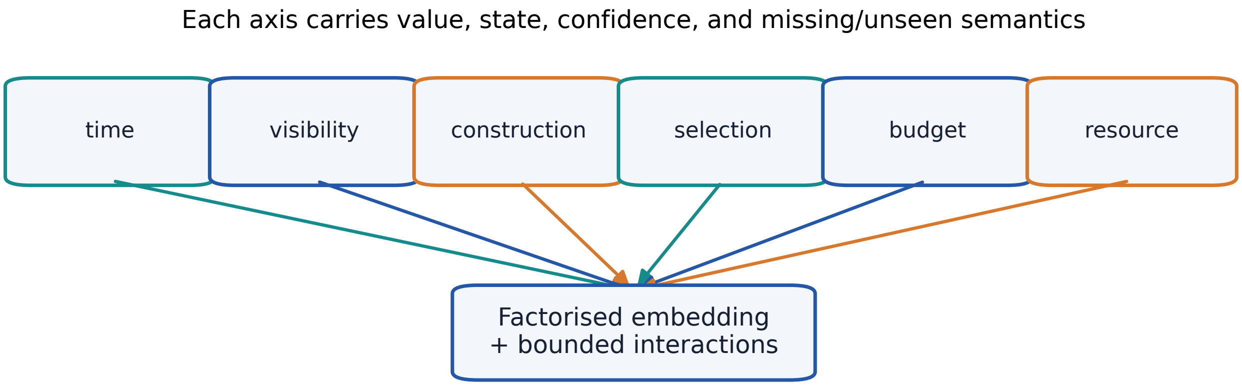

Figure 3: Factorised contract representation. Each axis preserves value, missing/unseen state, and confidence before bounded composition.

share an unknown token while retaining their state. The axis vectors h_j are composed as

$$h_c = \mathrm{MLP}\big([h_1; \dots; h_6; \{\mathrm{clip}(h_j^\top h_k)\}_{j<k}]\big)\,.$$

Bounded pairwise interactions are optional. An attention-over-axes variant, a flat contract MLP, a protocol one-hot memorization baseline, and a no-contract baseline share the output interface. Continuous means and scales are fitted on source contracts only. Encoder configuration, vocabulary, statistics, and interaction switches are serialized in each run manifest.

4.2 Expert-aware diagnostics

Each expert token may contain its probability, confidence, entropy, margin, logit norm, pairwise disagreement, ensemble variance, a source-fitted calibration proxy, feature and representation drift, visible degree statistics, graph density, isolated-node fraction, neighborhood truncation, edge-perturbation sensitivity, expected latency and memory, availability, and diagnostic confidence. Every diagnostic declares inputs, target-label requirements, source fitting, missing semantics, and cost. Label-dependent target error or prevalence is offline-only and rejected as a router input.

Diagnostics complement contract metadata. The contract may state isolated access while score disagreement reveals that a particular graph expert is unstable; conversely, noisy metadata can be downweighted when label-free signals remain coherent. Target covariates are used only when the selection/access coordinate permits them.

4.3 Contract, instance, and hierarchical routing

The contract router produces logits $z_e^c = g_c(h_c, d_c)$, one distribution per contract. The instance router produces $z_e^x = g_x(x, d_e)$. The main hierarchical router uses

$$z_e = z_e^c + \delta_e(x, c), \qquad \delta_e = \rho \tanh(g_x(x, d_e)),$$

where ρ bounds correction magnitude. A threshold can sparsify corrections. Contract-only and instance-only modes are exact ablations. Stability penalizes within-contract variation and response to admissible metadata, score, and availability perturbations. Invocation accounting records every positive-mass expert or the executed sparse selection policy.

4.4 Resource-feasible masking and fallback

Availability, task compatibility, memory, latency, and invocation constraints are conjoined into mask m_e before softmax:

$$\tilde{z}_e = \begin{cases} z_e, & m_e = 1, \\ -\infty, & m_e = 0. \end{cases} \qquad \pi_e = \mathrm{softmax}(\tilde{z})_e.$$

Unavailable probability is exactly zero and feasible probability sums to one. If every $m_e = 0$, the implementation emits zero expert mass, selected-expert sentinel -1, and forced abstention. A declared cheapest-feasible or feature-safe fallback may be used only when it is itself feasible. The mask and reason codes are persisted.

4.5 Robust objective and abstention

For source contract groups, the configurable objective is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_r \mathcal{L}_{\text{regret}} + \lambda_w \mathcal{L}_{\text{CVaR}} + \lambda_b \mathcal{L}_{\text{budget}} + \lambda_s \mathcal{L}_{\text{stability}} \\ + \lambda_a \mathcal{L}_{\text{abstain}} + \lambda_u \mathcal{L}_{\text{uncertainty}} + \lambda_1 \mathcal{L}_{\text{sparsity}} + \lambda_c \mathcal{L}_{\text{compute}}.\end{aligned}$$

Task, regret, and CVaR terms use source labels; deployment never does. Contract regret compares to one feasible expert after whole-contract risk aggregation. Budget surrogates are group-local and do not claim exact optimization of discontinuous Recall@K. Coefficients and source-validation thresholds are frozen before any target result is unlocked.

The abstention head predicts rejection from contract and label-free diagnostics. Source validation selects a threshold subject to a coverage/abstention capacity, then target inference applies the frozen decision. Forced all-unavailable abstention overrides it. Threshold perturbations are included in stability analysis.

4.6 Training and inference algorithms

Training. (1) Construct leakage-audited source graph views. (2) Fit continuous normalization, category vocabulary, source calibration proxies, and baselines on source train/validation only. (3) Compute expert scores and label-free diagnostics. (4) Compose feasible masks. (5) optimize the enabled objective terms over source contract groups. (6) Select abstention and model checkpoints on balanced source validation. (7) Freeze hashes, configuration, and thresholds before target scoring.

Inference. (1) Validate contract and resource status. (2) Encode supplied metadata and optional experimental inferred factors. (3) score admissible experts and diagnostics. (4) compute contract prior and bounded instance correction. (5) apply the mask before normalization. (6) force abstention if empty; otherwise route and apply the frozen threshold. (7) emit scores, weights, mask, selected expert, abstention, cost, and explanation record.

The optional latent contract-discovery encoder consumes graph statistics, feature moments, disagreement, drift, confidence distributions, and resource signals. It emits latent factors, uncertainty, and out-of-support score. It is disabled by default and compared with supplied, inferred, hybrid, and no-contract variants.

Complexity. With E experts, d router width, and $A = 6$ axes, contract composition costs $O(A^2 d)$ with pairwise interactions and $O(A^2 d)$ for one-head axis attention. Routing costs $O(Ed)$ per instance, excluding expert inference. Mask composition is $O(E)$. Resource reporting separates this overhead from expert latency.

5 Theory

The results below explain when contract conditioning is necessary and when factorisation can extrapolate. They do not establish empirical performance. Full proofs, counterexamples, and status labels appear in the supplement.

Theorem 1 (Fixed-mixture impossibility)**.** *Suppose two experts have risks $(0, \delta_1)$ on one contract and $(\delta_2, 0)$ on another, with $\delta_1, \delta_2 > 0$, and randomized selection risk is affine in mixture mass. Every contract-independent mixture has worst-contract regret at least $\delta_1\delta_2/(\delta_1 + \delta_2) > 0$; a contract-aware selector has zero regret.*

The two regrets are $(1 - w)\delta_1$ and $w\delta_2$; equalizing them gives the minimax value. The conclusion extends to resource-specific feasible sets only if their feasible optima still reverse. It need not hold for nonlinear prediction averaging without additional assumptions.

Proposition 1 (Regret decomposition)**.** *Insert intermediate policies that successively replace true contract representation, diagnostics, router, resource set, budget, abstention decision, and population risk with their deployed estimates. The resulting telescoping identity and triangle inequality bound regret by the sum of representation, diagnostic, routing, resource, budget, abstention, and finite-sample terms.*

Some terms are source-estimable or offline counterfactual; representation error may be theoretical. The decomposition is bookkeeping, not an assertion that every component is causally identifiable.

Figure 4: Theory intuition and its principal failure case. No real risk value is shown.

Table 2: Pre-registered benchmark layers. No dataset or result count is inferred here.

Layer	Family	Shift surface	Current status
Fraud	Elliptic	time / graph access / budget	saved outputs verified; pilot pending
Fraud	DGraphFin	time / graph access / resource	saved outputs verified; pilot pending
Controlled	fifteen mechanisms	isolated and correlated factors	tiny fixtures ready
Graph OOD	GOOD	official environments	not installed
Fallback	OGB molecular	scaffold / time	licence review pending

Theorem 2 (Compositional generalisation)**.** *Let* $R(a, c) = b(a) + \sum_j f_j(a, c_j) + h(a, c)$*, with* $|h| \le \epsilon_{\text{int}}$*. If source-observed axis effects have uniform errors* ϵ_j *and router optimization error is* ϵ_{route}*, then an unseen combination of those observed values has excess risk at most*

$$2 \sum_j \epsilon_j + 2\epsilon_{\text{int}} + \epsilon_{\text{route}}.$$

The proof applies the approximation bound to the learned and optimal actions. The result excludes unseen axis values. Arbitrary interactions can make ϵ_{int} large; an XOR construction has zero marginal effects and nonzero joint risk.

Proposition 2 (Feasible mask)**.** *Pre-softmax masking assigns exactly zero mass to unavailable experts and unit mass to a nonempty feasible set. Conjoining correctly measured memory, latency, and invocation predicates preserves those constraints. An empty set is represented explicitly rather than normalized.*

Proposition 3 (Selective-risk transfer)**.** *For a label-free acceptance event* A*, bounded loss* $[0, L]$*, density ratio* $dP_t/dP_s \le \kappa$ *on* A*, and target coverage at least* $\gamma > 0$*,*

$$R_t^{\text{sel}} \le \min\left\{L, \frac{\kappa P_s(A) R_s^{\text{sel}}}{\gamma}\right\}.$$

This is not distribution-free: unbounded density ratio, target-label-dependent acceptance, or vanishing target coverage invalidates the guarantee.

6 Three-layer benchmark

Layer A: fraud deployment shifts. The first saved-output pilot uses Elliptic and DGraphFin under strict inductive, isolated inductive, and transductive-structure protocols. Each protocol is mapped to explicit time,

Table 3: Canonical saved-output expert set for the first fraud pilot.

Expert	Information	Resource sensitivity	Evidence status
Feature MLP	node features	low graph dependence	members verified; result pending
GCN	features + visible graph	graph / memory / latency	members verified; result pending
GraphSAGE	features + sampled graph	graph / sampling / latency	members verified; result pending

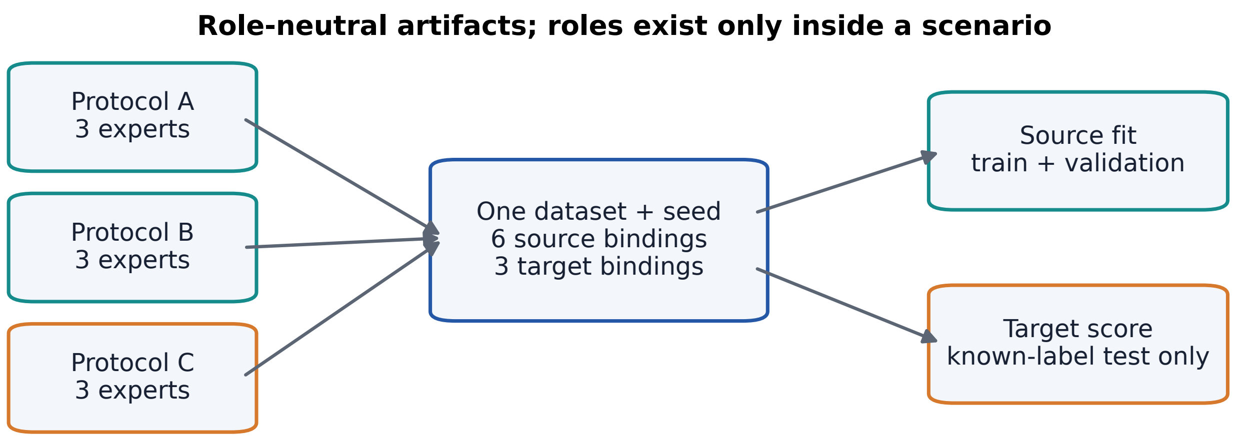

Figure 5: Scenario-local roles. A base artifact may change role across scenarios but cannot be both source and target inside one scenario.

visibility, and construction axes rather than treated as an opaque name. For each dataset, target protocol, and seed, the target protocol is held out and the remaining two provide six source expert bindings; the target provides three test bindings. The resulting 180 role-neutral coordinates, 60 scenarios, and 540 bindings are registered. All six canonical archives and 180 members pass checksum, schema, coordinate, chronology, label-known, cross-expert alignment, and cross-protocol row-scope audits; the pilot itself has not run.

Future fraud conditions add temporal drift, graph construction/visibility changes, expert unavailability, review-budget contraction, latency/memory limits, and combined shifts. Provider task units and label mappings are retained. Unknown labels are excluded; chronological and endpoint scope must validate before scoring.

Layer B: controlled mechanisms. Fifteen deterministic generators isolate feature corruption, edge deletion, degree shift, homophily reduction, heterophily reversal, temporal concept drift, delayed labels, neighborhood truncation, expert unavailability, review-budget contraction, correlated graph/feature shift, noisy metadata, missing axes, dynamic resources, and universal expert degradation. Every mechanism declares source and held-out contracts, qualitative router direction, pilot/full sizes, seeds, metrics, failure modes, and deterministic hash. Tiny fixtures test semantics; they do not claim fraud realism.

Layer C: non-fraud graph OOD. GOOD is the primary recognized family because it exposes official graph OOD environments and task-compatible node settings [5]. An OGB molecular scaffold/time family is a fallback subject to dataset-specific licence review and a graph-level expert bridge. No official code or dataset is installed in this build. Both adapters have tiny stand-ins that test schema, label access, and split handling but cannot support an official result claim.

Interventions and leakage controls. The benchmark includes held-out combinations, unseen values, noisy/partial metadata, counterfactual contracts, resource and budget masks, all-expert failure, and all-unavailable cases. Contract changes never mutate fixed expert scores during counterfactual routing tests. Dataset, protocol, expert, seed, fold, row ID, provider labels, scores, split, and label-known status must align before evaluation.

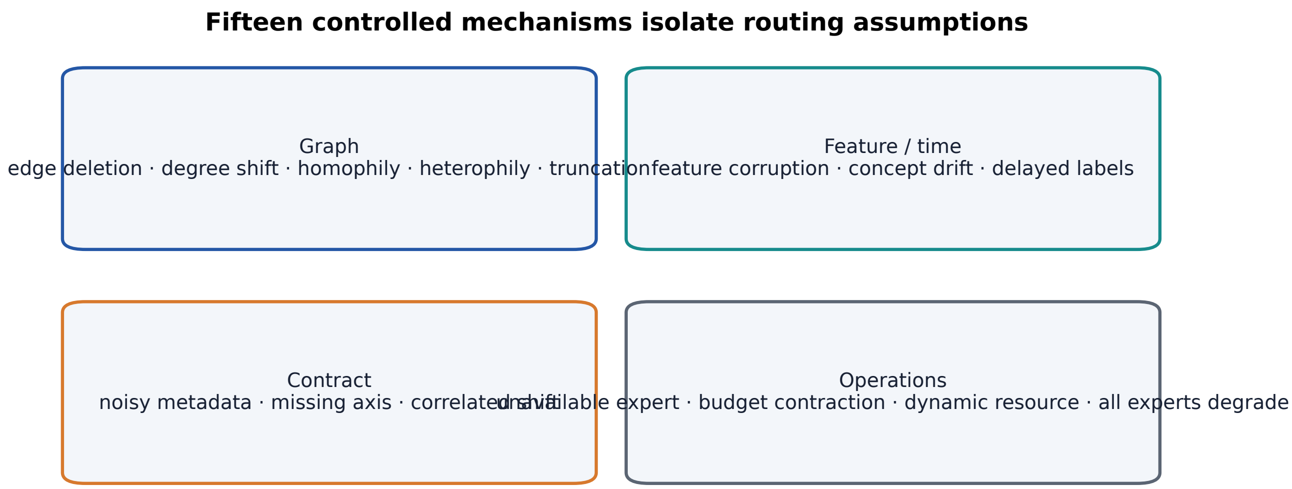

Figure 6: Controlled mechanism taxonomy. Expected directions are hypotheses, not hard-coded empirical conclusions.

Table 4: Baseline families and honest implementation status. Oracles are diagnostic only.

Family	Examples	Status	Headline eligibility
Simple	fixed, average, confidence, logistic/MLP gates	implemented internal	yes
Resource-aware	cheapest feasible, heuristic gate	implemented internal	yes
Graph MoE	Mowst	official available, not installed	blocked parity
Graph OOD	GraphMETRO, CIGA, EERM, GOOD	mixed licence/integration status	task-valid cells only
Robust	GroupDRO, VREx	internal objective variants	yes with common access
Oracle	whole-contract, instance-clairvoyant	implemented diagnostic	no

7 Experimental protocol

Experts and baselines. The V5 saved-output fraud pilot uses feature MLP, GCN, and GraphSAGE predictions for ten expert-training seeds. Its frozen primary family contains exactly four methods: uniform average, best fixed expert selected on source validation, a source-trained logistic gate, and CoReGraph. Whole-contract and instance oracles are offline diagnostics and cannot pass the deployment gate. Broader benchmark phases may evaluate Mowst-style graph mixtures, GraphMETRO, GroupDRO, VREx, CIGA, EERM, temporal models, protocol-valid adaptation, and resource-aware routing, but those methods are not silently added to the V5 pilot family. Official repositories remain separately pinned and uninstalled; unavailable licence or task-incompatible cells are reported, never approximated under official names.

Tuning and seeds. All deployable hyperparameters, normalization, diagnostics, checkpoints, and abstention thresholds use source train/validation only. The target protocol is absent from fitting. A target-unlabeled bundle contains predictions and permitted diagnostics but has no label field; the offline evaluator opens target labels only after a checksum-bound policy-freeze record exists. Seeds 1–10 index immutable expert predictions; router randomness is recorded but not counted as an independent dataset replication. Same-numbered seeds across datasets are never paired.

Outcomes. Primary outcomes are contract regret, maximum and CVaR regret, AUPRC, recall at fixed review budgets, selective risk, coverage, routing stability, expert invocation count, warmed inference and router latency,

Table 5: Primary fraud results. Every cell remains blocked until validated runs and frozen inference pass.

Method	Elliptic AUPRC	DGraphFin AUPRC	Mean regret	Worst regret
Uniform average	PENDING	PENDING	PENDING	PENDING
Best fixed expert	PENDING	PENDING	PENDING	PENDING
Source logistic gate	PENDING	PENDING	PENDING	PENDING
Strongest feasible official baselines	PENDING	PENDING	PENDING	PENDING
CoReGraph	PENDING	PENDING	PENDING	PENDING

Table 6: Worst-contract and CVaR outcomes.

Method	Maximum regret	CVaR regret
Uniform average	PENDING	PENDING
Best fixed expert	PENDING	PENDING
CoReGraph	PENDING	PENDING

peak memory, cost, and abstention rate. Regret always names its loss. Resource-blocked outputs remain explicit missing cells.

Statistical analysis. The paired unit is dataset, held-out target contract, and expert-prediction seed. Exact Wilcoxon and paired permutation tests, raw and robust standardized effects, and seed-block bootstrap intervals are computed separately per dataset. Holm correction applies within frozen claim families. A hierarchical dataset-seed bootstrap is secondary and cannot override a contradiction on one fraud dataset. Missing cells are not imputed.

Ablations and stress tests. Encoder ablations compare protocol one-hot, flat MLP, factorisation, bounded interactions, axis attention, uncertainty, discovery, and hybrid variants. Routing ablations compare contract, instance, and hierarchical modes. Diagnostic, regret/CVaR, budget, stability, uncertainty, sparsity, abstention, and mask terms are switched independently. Noisy categorical/continuous axes, delayed resource status, incorrect protocol labels, corrupted diagnostics, and out-of-support contracts are pre-registered.

Resource measurement and claim gates. Parameter count, FLOPs/proxy, expert and router latency, memory, throughput, invocation, review use, batch, warmup, hardware, and software are measured under eight resource profiles. The pilot cannot pass unless integrity/leakage gates are complete, worst-contract regret improves directionally on both fraud datasets, no catastrophic primary reversal occurs, masks are exact, abstention has useful nonzero coverage, effects are not one-seed artifacts, and no oracle feature enters deployment. Exact numerical thresholds may not be chosen from target results.

8 Results: blocked structure

Primary comparison. RESULT_PENDING[R1-PRIMARY-FRAUD] This paragraph will report per-dataset AUPRC and whole-contract Brier regret only after 180 prediction members, 60 scenarios, 540 bindings, paired completeness, and Holm-adjusted claim gates pass. No expected winner is stated.

Unseen compositions. RESULT_PENDING[R2-COMPOSITION] The protocol one-hot, flat MLP, factorised, interaction, and attention encoders will be compared on held-out combinations and, separately, unseen values. The composition theorem does not license an empirical claim.

Diagnostics and hierarchy. RESULT_PENDING[R3-ROUTING] Contract-only, instance-only, and hierarchical routing will be compared jointly on regret and stability. Expert-aware diagnostics must beat contract-only metadata under their frozen gate before the corresponding claim is permitted.

Robustness and abstention. RESULT_PENDING[R4-ROBUST-SELECTIVE] Mean-risk, regret, worst/CVaR, and abstention variants will report coverage with selective risk. Zero coverage is not a successful result.

Table 7: Contribution-isolating ablations.

Variant	Composition gap	Stability
Protocol one-hot	PENDING	PENDING
Flat contract MLP	PENDING	PENDING
No diagnostics	PENDING	PENDING
Contract-only / instance-only	PENDING	PENDING
No robust regret / no abstention	PENDING	PENDING
Full CoReGraph	PENDING	PENDING

Table 8: Failure analysis populated only from validated manifests.

Failure family	Incidence	Consequence
Wrong expert / all experts poor	PENDING	PENDING
Routing or abstention collapse	PENDING	PENDING
Resource or budget collapse	PENDING	PENDING
Metadata ambiguity / correlated shift	PENDING	PENDING

All result figures remain unpopulated templates. Their generation script accepts only validated result manifests and never plausible placeholder values. Current blocked claims include superiority, reduction in regret, useful coverage, cross-dataset consistency, resource efficiency, and latent-discovery benefit.

9 Failure analysis

Failure labels are frozen before results: wrong expert selected; all experts poor; overconfident diagnostics; routing instability; correlated shift; latent-contract ambiguity; budget collapse; resource-mask collapse; abstention collapse; source overfitting; baseline unfairness; and metric disagreement. Every evaluated row may carry multiple labels with evidence links.

Wrong selection is measured against the whole-contract feasible oracle after target labels unlock for evaluation. It is distinct from universal expert failure, where no route solves the contract. An empty feasible set is a resource event and must abstain; assigning mass anyway is a mask failure. Zero coverage, near-universal acceptance under high uncertainty, or threshold sensitivity are abstention failures rather than favorable selective-risk outcomes.

Counterfactual tests hold expert scores fixed while changing one resource, budget, visibility-metadata, cost, or uncertainty input. Unavailable mass must become zero. Tighter budgets may favor cheaper policies, and higher uncertainty may increase abstention when configured, but no directional performance conclusion is hard-coded where theory does not guarantee it. Metric disagreement is reported rather than resolved post hoc by selecting the most favorable outcome.

RESULT_PENDING[R5-FAILURES] Incidence, examples, and consequences remain blocked until validated predictions and failure manifests exist.

10 Resource and budget tradeoffs

Resource profiles cover all experts, one graph expert unavailable, all graph experts unavailable, tight memory, tight latency, tight review budget, combined graph/resource shift, and dynamic availability. The feasible set is constructed before routing from compatible measurement units. Estimated quantities retain an estimated status; OOM is ‘RESOURCE_BLOCKED‘ with attempted batch and envelope.

We measure warmed expert inference and router latency separately, as well as parameter count, FLOPs or a justified proxy, peak CPU/GPU memory, throughput, invocation count, review use, batch size, hardware, and software. Training runtime is excluded from inference-latency claims. Utility-resource frontiers will show missing and blocked cells instead of interpolation.

RESULT_PENDING[R6-RESOURCE] No latency, memory, throughput, cost, or utility number has been measured for Level-4. Consequently the paper makes no efficiency claim.

Table 9: Resource tradeoffs. No training runtime is substituted for latency.

Condition	Utility	Latency	Peak memory
All experts available	PENDING	PENDING	PENDING
Graph experts unavailable	PENDING	PENDING	PENDING
Tight latency / memory	PENDING	PENDING	PENDING
Dynamic availability	PENDING	PENDING	PENDING

Table 10: Noisy and partial contract robustness.

Intervention	Regret	Abstention
Missing axis	PENDING	PENDING
Noisy categorical axis	PENDING	PENDING
Noisy continuous axis	PENDING	PENDING
Corrupted diagnostics	PENDING	PENDING
Out-of-support contract	PENDING	PENDING

11 Limitations and ethics

Scientific limitations. Deployment contracts compress organizational, data-collection, and infrastructure choices into six axes. Important latent factors may be absent, supplied metadata may be wrong, and arbitrary interactions defeat factorisation. The expert set bounds the attainable policy; a feasible oracle is not a population optimum. Diagnostics can drift or be strategically manipulated. Source-calibrated abstention has no distribution-free target guarantee. Resource feasibility depends on accurate, comparable measurements.

The first real layer contains only two fraud datasets and three inherited experts. Elliptic and Elliptic++ are correlated, IBM AML is synthetic, T-Finance timestamp semantics require audit, and provider licences constrain redistribution. A recognized graph-OOD family helps test transfer but cannot establish universal graph generalisation. Official baseline parity may remain incomplete where licences, dependencies, task units, or compute prevent a faithful comparison. Such cells restrict claims rather than disappearing.

Financial and societal risks. Fraud scores can affect access to financial services and trigger costly investigation. False positives can burden individuals and review teams; false negatives can expose institutions and customers to harm. Labels may encode historical enforcement and representation bias. Aggregate performance can conceal group-specific burdens, while graph relations can proxy sensitive associations.

CoReGraph is intended as decision-support research, not autonomous adjudication. A human review budget is an explicit contract rather than permission to deny service automatically. Deployments require domain governance, legal review, privacy/security controls, subgroup and error-burden analysis where lawful, appeal and correction processes, monitoring for drift and gaming, and conservative action when metadata or resources are uncertain. The artifact contains no provider records or personally identifying data.

Research integrity and related work. All empirical cells are blocked until evidence passes. Missing licences and measurements are labelled. The related FraudShiftBench manuscript is disclosed; its 249 scientific files remain unchanged, and its results are not repurposed as evidence for CoReGraph. Any LLM-use disclosure and target-year venue policy must be checked before submission; authors retain responsibility for every claim and citation.

12 Conclusion

We formulated graph deployment as compositional generalisation over typed contracts rather than selection under one protocol name. CoReGraph joins factorised uncertain metadata, expert diagnostics, hierarchical routing, exact resource masks, robust regret, and controlled abstention. The theory states why fixed mixtures can fail and when composition, masking, and selective transfer admit scoped guarantees. The benchmark and analysis plan are frozen across fraud, controlled mechanisms, and graph OOD.

The engineering and scholarly structure do not substitute for results. The six canonical saved-output archives and their 180 prediction members are locally checksum-verified. A checksum-gated V5 executor now materialises

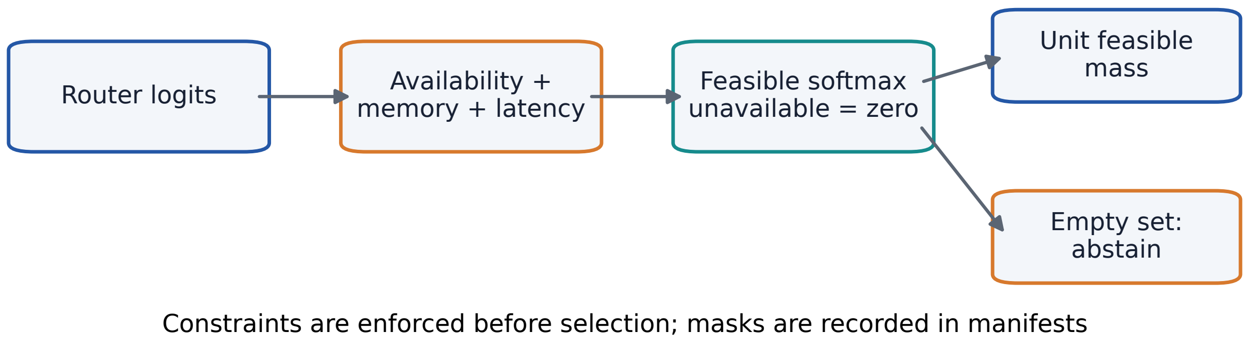

Figure 7: Resource-feasible masking. The empty-set branch is an explicit abstention state.

the frozen 60 scenarios and 540 bindings, enforces a target-label firewall, freezes each policy before offline evaluation, resumes from checksum-bound coordinates, and completes a deterministic 240-coordinate synthetic campaign. The real saved-output pilot has not run, official baselines are not installed, resources are unmeasured, and every numerical paper claim remains blocked. The next scientific step is a separately authorised real pilot using the frozen specification; no empirical success is concluded here.

Reproducibility statement. The anonymous artifact specifies contract schemas, scenario bindings, leakage controls, run manifests, source-only tuning, statistical gates, and deterministic tiny checks. Provider data, prediction payloads, credentials, private paths, and unresolved-licence outputs are excluded. The manuscript is explicitly `RESULTS_BLOCKED`.

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